

Article

Multicomponent Adsorption of Paracetamol and Metronidazole by Batch and Fixed-Bed Column Processes: Application of Monte Carlo Bayesian Modeling

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Abstract

This study addresses the growing concern of water contamination by pharmaceutical residues, focusing on the simultaneous removal of paracetamol (PAR) and metronidazole (MTZ). Batch and fixed-bed column adsorption processes were evaluated using activated carbon. In the batch experiments, the effects of pH (3, 7, and 11), adsorbent mass (0.5, 1.25, and 2 g), and contact time (10, 30, and 60 min) were evaluated, while the fixed-bed column was optimized considering initial pollutants concentration (30, 40, and 50 mg/L), adsorbent mass (0.5, 0.75, and 1 g), and flow rate (5, 10, and 15 mL/min) to improve the maximum adsorption capacity of the bed for both pollutants (q_{maxPAR} and q_{maxMTZ}). Parameter estimation and model selection were performed using a Bayesian Monte Carlo approach. Optimal conditions in the batch system (pH = 7, $W = 2$ g, and time = 60 min) led to high removal efficiencies for both compounds ($\geq 98\%$), while in the column system, the initial pollutant concentration was the most significant parameter to improve the maximum adsorption capacity of the bed, resulting in values equal to 49.5 and 43.6 mg/g for PAR and MTZ, respectively. The multicomponent Gompertz model showed the best performance for representing the breakthrough curves and is suitable for scale-up ($R^2 \geq 0.75$). These findings highlight the complexity of multicomponent adsorption and provide insights, contributing to the development of more efficient and sustainable water treatment technologies for pharmaceutical residues.

Keywords: water contamination; pharmaceutical residues; adsorption; batch process; fixed-bed column; Bayesian technique



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1. Introduction

Although pharmaceuticals are essential for treating various medical conditions, water contamination by pharmaceutical residues, such as paracetamol (PAR), an analgesic [1], and metronidazole (MTZ), widely used as an antibiotic and antiparasitic [2], has become a growing environmental and public health concern. These compounds are increasingly common in water bodies due to inadequate discharge of domestic, hospital, and industrial effluents and the lack of effective wastewater treatment [3,4]. Even at low concentrations, both PAR and MTZ can cause adverse impacts on aquatic ecosystems and pose potential risks to human health due to their presence in an unmetabolized form and challenges in wastewater management [5,6]. Therefore, water treatment strategies must be developed and

improved to address the challenges posed by the presence of these emerging contaminants, ensuring water security and public health.

Regulatory agencies have proposed threshold values for the presence of pharmaceutical residues in aquatic environments. For instance, the European Medicines Agency (EMA) and various national agencies have suggested environmental quality standards (EQS) in the range of 0.01 to 10 µg/L for pharmaceutical residues [7]. Specifically, proposed maximum concentrations for PAR and MTZ in surface or drinking water are typically below 1–2 µg/L, although exact values may vary depending on the country or guideline [8]. In the United States, the Environmental Protection Agency (EPA) includes several pharmaceutical compounds in its Contaminant Candidate List (CCL), and although not all have enforceable limits, both are monitored due to their potential environmental relevance [9]. In contrast, Brazil currently does not establish maximum permitted concentrations for these specific pharmaceutical compounds in its water quality regulations, such as CONAMA Resolution No. 357/2005, highlighting a regulatory gap that further underscores the need for improved monitoring and treatment approaches.

For many years, the predominant approach in water and effluent analysis and treatment focused on individual compounds, treating each contaminant separately [2,10–12]. This results in a limited understanding of the potential synergistic or antagonistic effects among different substances present in the water [13]. As a result, treatment methods tended to be specific to a single contaminant, neglecting the complexity of contamination by multiple pollutants.

The simultaneous removal of contaminants emerges as a pressing need due to the complexity of water contamination by substances from various sources [14–16]. However, this approach faces significant technical challenges since these contaminants can interact with each other, affecting the efficacy of removal processes and potentially causing adverse impacts. Furthermore, compliance with regulations and standards regarding concentration limits in water requires the development of more efficient and economical treatment technologies capable of dealing with this complexity in an integrated manner [14,17].

There are some studies dedicated to developing processes aimed at the simultaneous removal of contaminants in water [18–22]. Among these, those exploring the principles of adsorption stand out [23–27]. Being a technique with benefits such as high efficiency and easy implementation in treatment processes, adsorption is widely employed for the removal of different contaminants, applicable in batch processes suitable for treating small water volumes or fixed-bed columns preferable for larger volumes due to their efficiency and continuous treatment capability [28–30]. Additionally, the use of conventional adsorbents like activated carbon offers a unique combination of properties, including large surface area and porosity, favoring the effective adsorption of pollutants [31].

The adsorption models can be characterized as an inverse problem since they are non-linear equations. In other words, the inverse problem in adsorption models arises when we aim to estimate the model's parameters (such as adsorption rate, adsorption capacity, and others) using observed data. Essentially, this involves “inverting” the process to determine these parameters from the known outcomes. Different approaches can be used to determine the parameters of adsorption models, these being linear regression [32,33]; non-linear regression through the minimum square method [34]; and, more recently, non-linear regression applying Bayesian statistics [35,36]. These studies show that the Bayesian technique has an upfront advantage when it comes to estimate the values of models due to the flexibility in modeling, incorporating prior knowledge and information as a prior distribution, consequently, providing more reliable and statistically accurate results [37].

In this context, considering the increasing complexity of water contamination and the identified gaps in the literature, the main objective of this study is to investigate the

effectiveness of simultaneous removal of paracetamol and metronidazole using batch and fixed-bed column adsorption processes. This research aims to provide valuable insights for the development of more efficient and sustainable technologies for treating water contaminated by pharmaceutical residues.

2. Materials and Methods

2.1. Pharmaceuticals and Adsorbent

The pharmaceuticals paracetamol (CAS 103-90-2) and metronidazole (CAS 443-48-1), with analytical purity of 98% and >98%, respectively, were obtained from Sigma-Aldrich (St. Louis, MO, USA). Both compounds were selected as adsorbates for this study because of their widespread occurrence as pharmaceutical contaminants in water sources, as well as their persistence in the aquatic environment and potential adverse effects on both aquatic ecosystems and human health. Solutions were prepared immediately before use by appropriate dilution from stock solutions. Solutions of 0.1 M hydrochloric acid (HCl) or sodium hydroxide (NaOH), 98% purity, by Neon (Rio de Janeiro, RJ, Brazil) were used for pH adjustment. Granular activated carbon (CAS 7440-44-0), with particle size between 1.00 and 1.40 mm, was provided by Synth (Diadema, SP, Brazil). This solid was selected due to its proven effectiveness in removing emerging pollutants and its flexibility for use in various configurations, such as batch systems and fixed-bed columns. Its physical properties, including a high specific surface area and highly porous structure, combined with versatile chemical surfaces, facilitate efficient interactions with pharmaceutical molecules. Furthermore, its wide commercial availability and cost-effectiveness make it a practical and reliable choice.

2.2. Initial Characterization of the Adsorbent

The carbon activated was characterized by X-ray diffraction (XRD) using a Bruker D2 Phaser diffractometer (Bruker AXS GmbH, Karlsruhe, Germany) with an angular range of 5–90° (2 θ), CoK α radiation (λ = 1.542 Å), and an exposure speed of 0.05° 2 θ s^{−1}. Physical adsorption/desorption tests of N₂ at 77 K were also performed, and the specific surface area was calculated by the BET method, while the pore volume and diameter were calculated using the Barrett–Joyner–Halenda (BJH) equation. Fourier-Transform Infrared Spectroscopy (FTIR) was obtained with a Nicolet 6700 spectrophotometer over a wavelength range of 4000–400 cm^{−1} at room temperature, with a resolution of 4 cm^{−1}, in 16 scans.

2.3. Analytical Determination of Pharmaceutical Compounds

The concentration of the pharmaceuticals, both in the initial solution and in the adsorbed phase, was determined using ultraviolet–visible (UV-vis) spectrophotometry with the KASUAKI equipment, model IL-0082-Y (KASUAKI, Recife, Brazil). The detection ranges were set at 243 nm for paracetamol and 320 nm for metronidazole. The conversion from absorbance to concentration was performed using Equation (1), with the linear coefficient derived from the calibration curve for each pharmaceutical.

$$C = \frac{A}{a} \quad (1)$$

where C is the solute concentration (mg/L); A is the absorbance at the respective wavelengths (nm); and a is the linear coefficient of the calibration curve.

To assess potential peak overlap and interferences in the adsorption signals during the analytical determination of the pharmaceutical compound mixture evaluated in this study, an initial solution scan was conducted over a wavelength range of 190–590 nm.

2.4. Procedure and Experimental Design

2.4.1. Batch Adsorption

In this study, batch adsorption was optimized using a minimal number of experimental runs through the Box–Behnken design (BBK) and response surface methodology (RSM). STATISTICA 10.0® software was utilized for this purpose. The effects of the variables and their interactions were assessed using analysis of variance (ANOVA). RSM was employed to predict the optimal conditions based on the F-value and p-value. The experimental design considered three factors: pH (x_1), adsorbent mass (x_2), and contact time (x_3), each evaluated at three levels, resulting in a total of 15 experiments (see Table 1). Two response variables were selected: the removal efficiency of the compounds (R%—Equation (2)) and the adsorption capacity of the adsorbent (Q_e —Equation (3)).

$$R\% = \frac{C_i - C_e}{C_i} \times 100 \quad (2)$$

$$Q_e = \left[\frac{(C_i - C_e) \times V}{m} \right] \quad (3)$$

where C_i is the initial concentration of the pharmaceuticals (mg/L), C_e is the equilibrium concentration (mg/L), Q_e is the adsorption capacity of the carbon activated (mg/g), V is the volume of the solution (L), and m is the mass of the carbon activated (mg).

Table 1. Values of the variables evaluated for the simultaneous batch adsorption of paracetamol and metronidazole.

Level	Variables		
	pH	Solid Mass (g)	Time (min)
(−1)	3	0.50	10
0	7	1.25	30
(+1)	11	2.00	60

The batch adsorption tests were carried out using 100 mL solution volumes. An initial solution containing a mixture of pharmaceuticals at a concentration of 20 mg/L was added to an Erlenmeyer flask, followed by the addition of the carbon activated solid, pH adjustment, and agitation at 150 rpm on an orbital shaker (New Lab, model NL 161-04, São Paulo, Brazil) maintained at a controlled temperature of 25 °C. This methodology is consistent with previous studies [38,39]. The process duration for each assay was determined according to the experimental design. The standard deviation was calculated based on the central point.

The second-order polynomial model (Equation (4)) was used to approximate the mathematical relationship between the independent variables.

$$Y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \sum_{j=1}^k \beta_{ij} x_i x_j + \sum_{i=1}^k \beta_{ii} x_i^2 + \varepsilon \quad (4)$$

where Y refers to the predicted response, while β_{ij} shows the interaction coefficient, β_{ii} represents the quadratic, β_i shows the linear coefficient, and β_0 is the constant coefficient. In addition, x_i and x_j represent the independent variables, and ε and k , respectively, refer to the residual error and the number of independent variables.

2.4.2. Fixed-Bed Column Adsorption

Similar to batch adsorption, fixed-bed column adsorption was performed using a 2^3 design, with duplicates at the central point (*), and analyzed with STATISTICA 10.0®

software. The effects of the variables and their interactions were examined through analysis of variance (ANOVA), while response surface methodology (RSM) was used to forecast the optimal conditions based on the F-value and *p*-value. The experimental design included three factors: the initial concentration of pharmaceuticals in the solution (C_0 , x_1), the bed mass (W , x_2), and the volumetric feed flow rate (Q , x_3), each evaluated at three levels, along with triplicates at the central point, resulting in a total of 11 experiments (see Table 2). The maximum adsorption capacity of the fixed-bed column was chosen as the response factor for this phase ($q_{\max}$ —Equation (6)).

$$C_{\text{sat}} = 0.95C_0 \quad (5)$$

$$q_{\max} = \frac{C_0 Q}{1000W} \int_0^{t_f} \left(1 - \frac{C_{\text{sat}}}{C_0}\right) dt \quad (6)$$

where C_{sat} is the solute concentration at saturation time (mg/L); C_0 is the initial concentration of pharmaceuticals at the column inlet (mg/L); $q_{\max}$ is the maximum adsorption capacity of the bed (mg/g); t_f (min) is the final time; W (g) is the mass of adsorbent applied to the bed; and Q (mL/min) is the volumetric flow rate of the column.

Table 2. Values of the variables evaluated for the simultaneous adsorption of paracetamol and metronidazole in a fixed-bed column.

Level	Variables		
	C_0 (mg/L)	W (g)	Q (mL/min)
(−1)	30	0.50	5
0	40	0.75	10
(+1)	50	1.00	15

The experiments were carried out using a borosilicate glass column with an internal diameter of 12 mm and a total length of 20 cm. The pharmaceutical solution was pumped upward through the column at a controlled flow rate using an ISMATEC peristaltic pump (model MCP), in accordance with the system used by Costa et al. [39]. The length of the bed, formed by adding granular activated carbon, varied according to the amount of adsorbent used. To prevent void spaces and preferential pathways, as well as the carryover of GAC particles by the treated solution, the bed was positioned between 6 g of glass beads and a filter paper with a grammage of 80 g/m² and a thickness of 205 µm (JProlab, São José dos Pinhais, PR, Brazil). Distilled water was pumped as the influent solution for at least 20 min to equilibrate the column prior to each operation. All experiments were conducted at 25 °C, using distilled water to dilute the pharmaceuticals from the stock solution.

Breakthrough curves were generated by continuously monitoring the pollutant concentration over time during the process. For simulating the breakthrough curve, mathematical models that simplify the mass balance, typically represented by a system of partial differential equations [40], were employed to facilitate potential process scaling and to identify the limiting step of the adsorption process while determining the kinetic parameters of the column. The modeling of multicomponent adsorption systems is complex, due to simultaneous effects and displacement of the curves, thus has been little studied in the literature. For modeling the multicomponent breakthrough curves the empirical models proposed and established by [40], these being the logistic (Equation (7)) and Gompertz (Equation (8)), were applied in this work.

$$\frac{C}{C_0} = \frac{1}{1 + \exp[k(\tau - t)]} + \frac{ck^* \exp[k^*(\tau^* - t)]}{\{1 + \exp \exp[k^*(\tau^* - t)]\}^2} \quad (7)$$

$$\frac{C}{C_0} = \exp \exp \{ -\exp \exp [k(\tau - t)] \} + ck^* \frac{\exp \exp \{ -\exp \exp [k^*(\tau^* - t)] \} \cdot \exp [k^*(\tau^* - t)]}{\exp \exp \{ -\exp \exp [k^*(\tau^* - t)] \}} \quad (8)$$

where C_0 (mg/L) and C (mg/L) are the influent and effluent concentrations, respectively; k (1/min), k^* (1/min), τ (min) and τ^* (min) are the adjustable parameters; and c is the dimensionless parameter, which reflects the strength of the displacement process.

The selection of the most appropriate model was carried out using the following statistical metrics: (1) coefficient of determination (R^2), commonly used in adsorption studies [39]; (2) adjusted coefficient of determination (R^2_{adj}), both based on maximization; (3) Akaike information criterion (AIC) [40,41]; (4) corrected Akaike information criterion (AICc) [42]; and (5) Bayesian information criterion (BIC) [43], with the last three being Bayesian metrics focused on its minimization.

2.4.3. Mathematical Modeling and Parameter Estimation

The mathematical modeling and parameter estimation in this study involves the application of Bayes' theorem, presented in Equation (9). This can be reformulated as Equation (10) for more practical application.

$$p(\theta|Y) = \frac{p(Y|\theta)p(\theta)}{p(Y)} \quad (9)$$

$$p(\theta|Y) \propto p(Y|\theta)p(\theta) \quad (10)$$

where $p(\theta|Y)$ is the posterior probability density function; Y is the data vector (in the present work is the ratio between C and C_0); θ is the vector of the parameters of interest; $p(Y|\theta)$ is the likelihood function; $p(\theta)$ is the prior probability density function; and $p(Y)$ is the function of the normalization constant, called the evidence.

The data in this work is considered independent and identically distributed. Thus, under these assumptions, the likelihood function can be obtained by applying Equation (11).

$$p(Y|\theta) = \frac{1}{(2\pi\sigma_T^2)^{\frac{-n}{2}}} \exp \left(-\frac{(Y - F(\theta))^T (Y - F(\theta))}{2\sigma_T^2} \right) \quad (11)$$

where σ_T^2 is the measurement uncertainty variance, also considered $0.01\max(C/C_0^{\text{experimental}})$, accounted for intrinsic sensor errors, as well as operational and measurement noise; F is the data calculated according to the parameters to be estimated; Y is the data applied to the models.

To implement the MCMC method, the input data used are listed in Table 3.

Table 3. Data input to apply the MCMC method.

NMCMC: states of the Markov chains	30,000
Burn-in period	16,000
w: search step	0.003
C.I.: credibility interval	99%
θ_0 : parameter initial guess	Based on the results found by Hu et al. [40] for binary adsorption

The a priori distributions of the parameters were defined based on the literature, considering typical values observed for adsorption systems with activated carbon [40]. Uniform distributions ($U(0,1)$) were chosen with parameters defined to reflect high prior knowledge based on the literature [44,45]. To ensure the robustness of the inference, the

convergence analysis of each parameter of the model was performed by Markov chains; the convergence of the posterior distributions, the autocorrelation, and the trace plots were checked; and, to build the chains and apply the MCMC method, the Metropolis–Hastings (MH) algorithm was used [45–47].

3. Results and Discussion

3.1. Initial Characterization of the Adsorbent

Figure 1 presents the textural characterization for the activated carbon used in the present study. The specific surface area and pore distribution of the materials were determined through N₂ adsorption/desorption. Subsequently, the specific values obtained from this characterization step are shown.

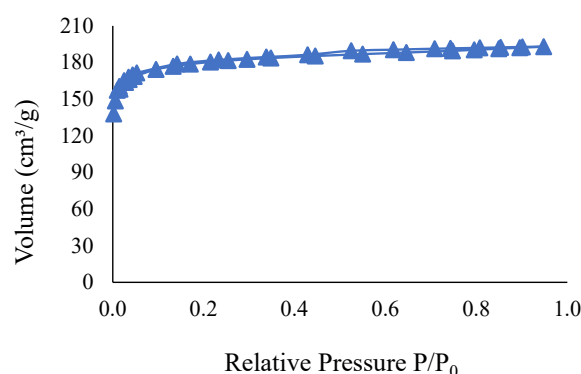


Figure 1. N₂ adsorption/desorption isotherms for the activated carbon.

As shown in Figure 1, the samples do not display a steep slope, indicating that adsorption does not significantly increase at high values of P/P_0 . Additionally, there is a trend toward stabilization of adsorption at these values. All curves exhibited distinct paths for adsorption and desorption, which is characteristic of hysteresis [48]. According to Costa and Féris [49], the presence of hysteresis suggests a tendency toward mesopore formation.

The specific surface area of the granular activated carbon used in this study was estimated at 582.60 m²/g, comparable to other commercial activated carbons reported in previous studies [36,37,46]. In terms of surface area, volume, and mean pore diameter, the sample exhibited values of 348.67 m²/g, 0.13 cm³/g, and 4.83 Å, respectively, indicating that it is suitable for microporous materials [48]. According to the classification by the International Union of Pure and Applied Chemistry (IUPAC), the isotherm curves displayed behavior characteristic of type I isotherms, which indicates micropore adsorption. Based on the empirical classification of hysteresis curves, the observed hysteresis can be classified as type H4, typical of materials with ordered and uniform pores, often found in complex materials that contain both micropores and mesopores.

The XRD profile of the adsorbent before the adsorption of pollutants is illustrated in Figure 2. Two peaks of graphite carbon diffraction can be observed in the reflections at angles of 2θ near 25° and 45°, indicating that an amorphous phase is formed, and the activated carbons are ordered by the stacking of micrographs [50].

Figure 3 shows the FTIR spectrum of the activated carbon before adsorption process, in terms of percentage of transmittance (%T) and wave number (cm^{−1}).

From Figure 3 it is possible to observe peaks between the 2300 and 2000 cm^{−1} bands and also small changes between the 1700 and 1500 cm^{−1} lengths. The 2360 and 1645 cm^{−1} peaks are indicative of carboxylic structures (C=O), while the 1629 cm^{−1} band is related to the presence of C=C in the aromatic rings, and 1558 cm^{−1} corresponds to the elongation of the C=O bonds, present in the structure of the solid. The absence of significant peaks

at $2800\text{--}3000\text{ cm}^{-1}$ may indicate little presence of aliphatic residues or adsorbed organic compounds, which is expected before the adsorption process [48].

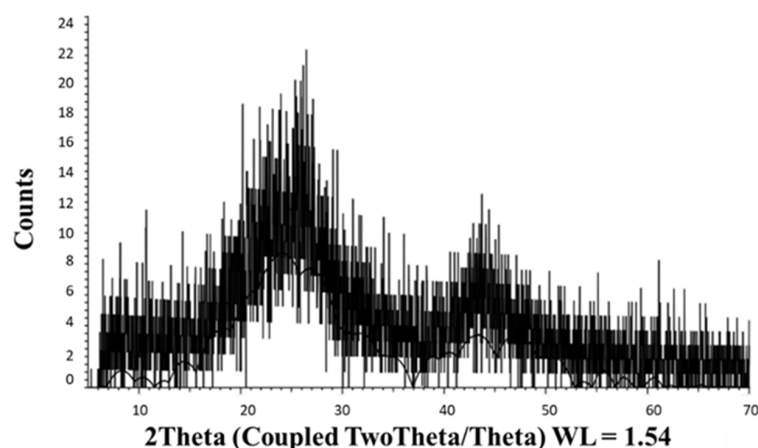


Figure 2. X-ray diffraction pattern of the carbon activated before the adsorption.

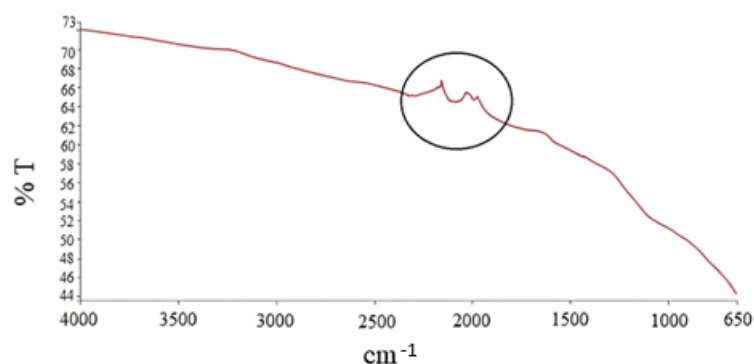


Figure 3. FTIR spectrum of the activated carbon before adsorption process.

3.2. Batch Adsorption Evaluation

Table 4 displays the experimental matrix along with the responses obtained for the simultaneous adsorption of pharmaceuticals in the batch process.

Table 4. Experimental matrix and responses for the simultaneous batch adsorption of pharmaceuticals.

Run Order	Independent Variables			Response	
	pH	Solid Mass (g)	Time (min)	Removal (%)	Q_e (mg/g)
1 *	7	2.00	60	98.1	4.0
2	3	1.25	60	97.5	3.4
3	3	0.50	30	76.3	2.7
4	3	2.00	30	96.1	2.1
5	11	2.00	30	93.6	1.6
6	7	1.25	30	85.8	1.4
7	7	1.25	30	83.4	1.3
8	7	1.25	30	89.7	1.6
9	7	0.50	10	56.7	2.1
10	11	0.50	30	72.2	2.5
11	11	1.25	60	97.3	3.2
12	11	1.25	10	71.1	1.1
13	3	1.25	10	72.7	1.2
14	7	2.00	10	77.1	0.8
15	7	0.50	60	86.2	3.1

Legend: * optimized results.

Figure 4 illustrates the response surface plots of the variable interactions derived from the applied experimental design.

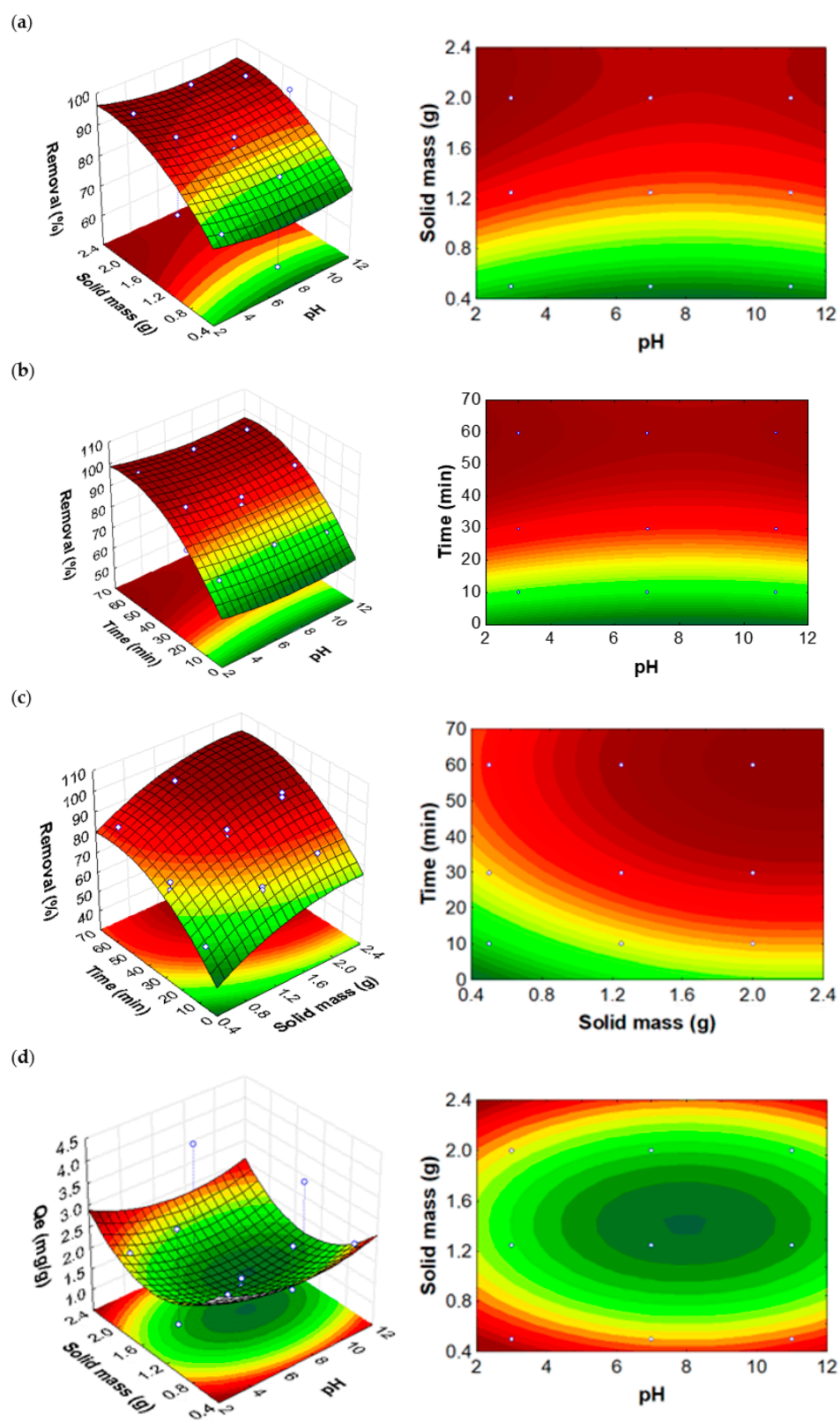


Figure 4. Cont.

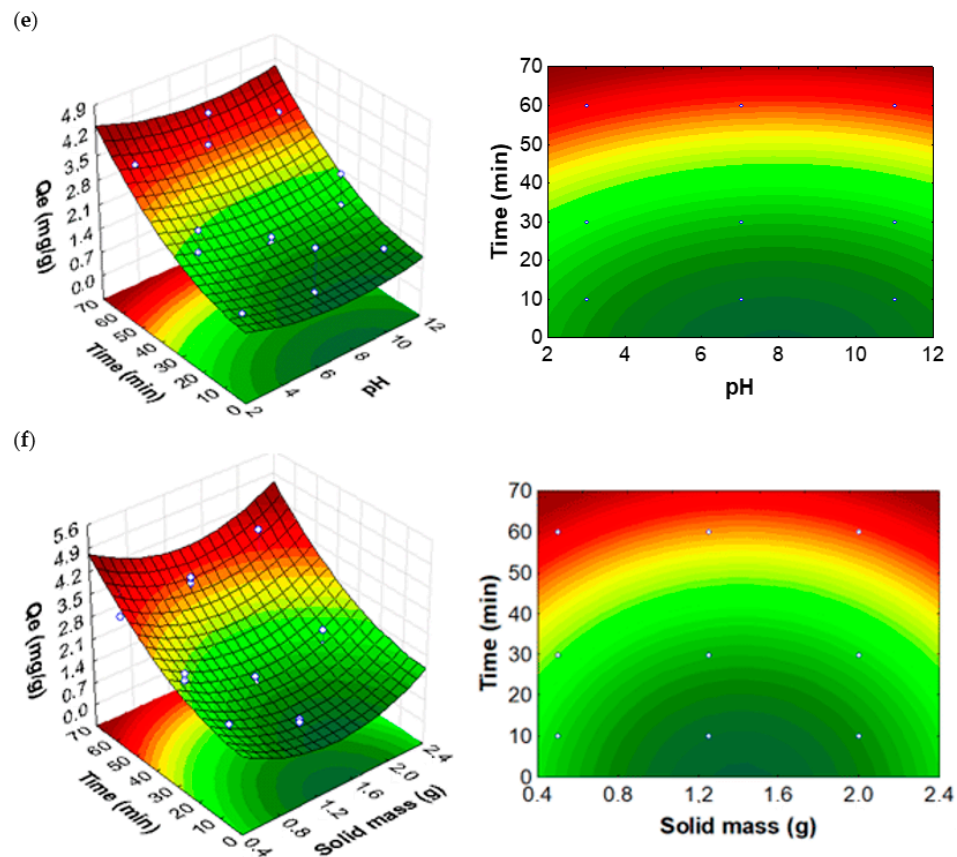


Figure 4. Surface area and contour plots for pharmaceuticals batch adsorption for %R: (a) solid mass and pH, (b) time and pH, (c) time and solid mass; Q_e (mg/g): (d) solid mass and pH, (e) time and pH, (f) time and solid mass. Red areas indicate maximum response, green shows minimum, and yellow represents intermediate values. The response is maximized in red zones.

The results indicate the impact of the independent variables on the effectiveness of the adsorption process. Overall, it appears that the pH effect does not lead to significant variations in the removal rates and adsorption capacity, suggesting an indirect relationship between the medium pH and the response variables (Figure 4a–e). The mass of the adsorbent does not show a clear trend concerning the adsorption capacity results (Q_e ; Figure 4d–f), revealing a non-linear relationship between mass and adsorption efficiency. However, when assessing the removal of pharmaceuticals, it is observed that the interaction between solid mass and solution pH (Figure 4a) results in the highest removal rates, with values approaching 98% as the amount of solid increases. A similar trend can be noted in the interaction between solid mass and process time (Figure 4c).

Costa et al. [39] investigated the adsorption of TC on commercial activated carbon similar to that used in this study. They found that the efficiency of TC removal increased gradually with the concentration of the adsorbent, reaching a maximum removal rate of 92.7% at a concentration of 3 g/L. Likewise, Jurado-Davila et al. [51] reported that the percentage of pharmaceutical removal rose with an increase in GAC concentration, with pollutant removal gradually increasing until equilibrium was reached. These findings can be attributed to the adsorption behavior of porous materials with a high specific surface area.

Contact time consistently affects all experiments, with a general increase in removal efficiency and adsorption capacity as contact time lengthens (Figure 5b,c,e,f). This behavior is expected and is characteristic of adsorption processes, where longer contact times facilitate more complete interactions between the adsorbent and adsorbate [24,52].

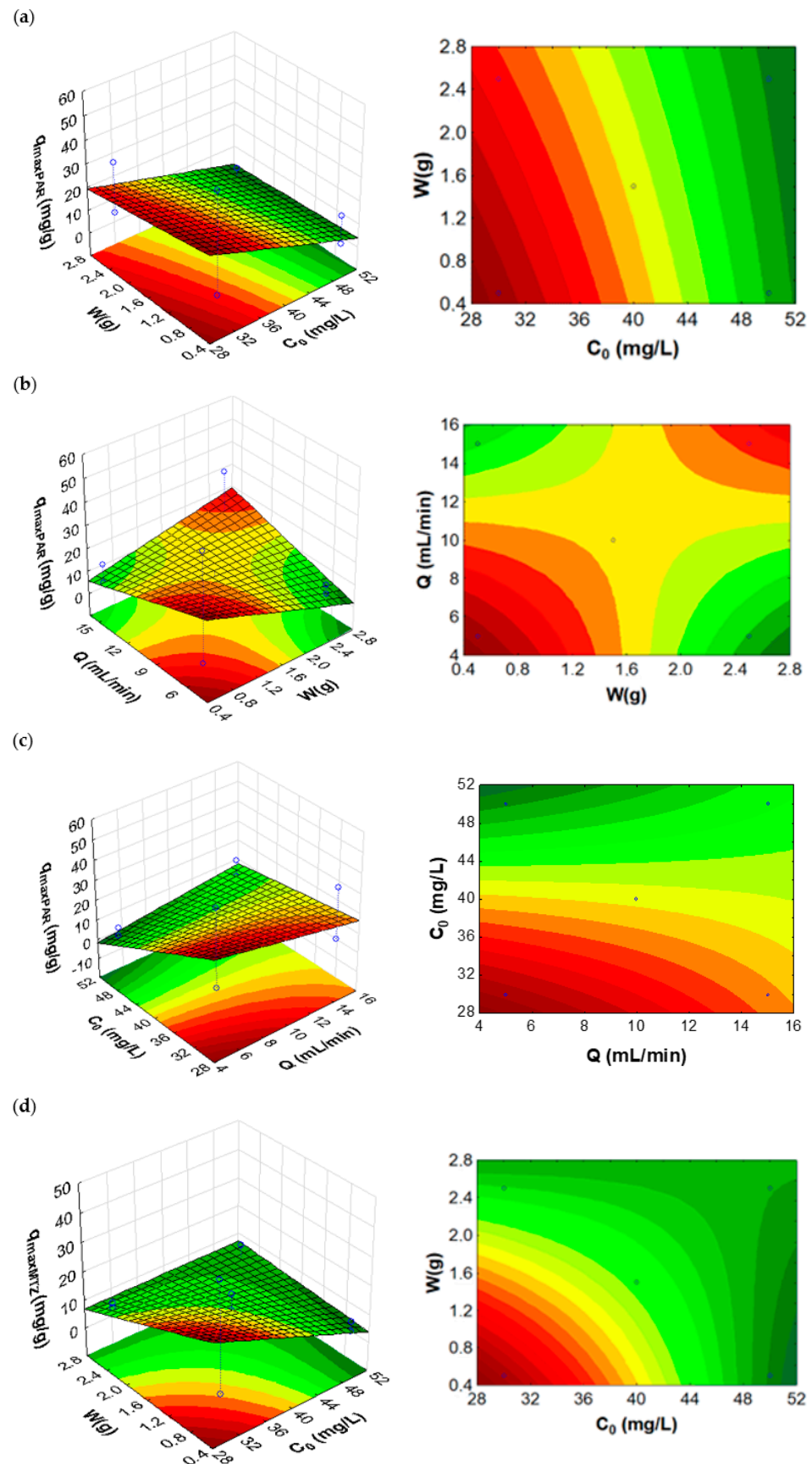


Figure 5. Cont.

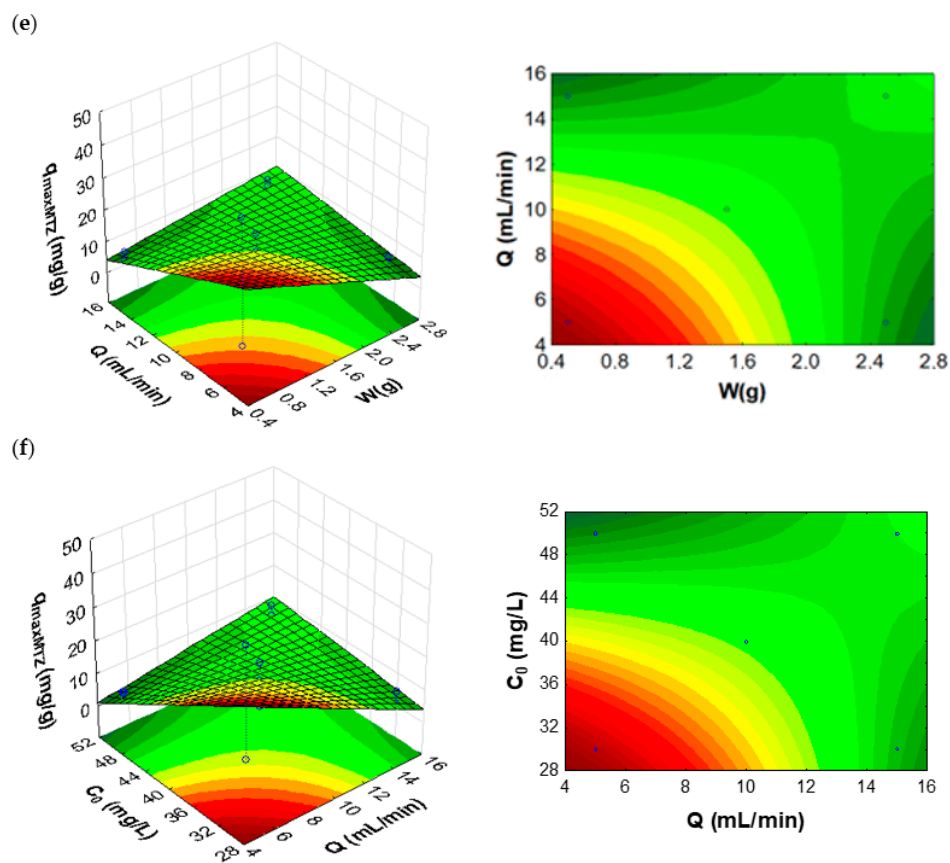


Figure 5. Surface area and contour plots for simultaneous pharmaceuticals adsorption by fixed-bed column for $q_{\max\text{PAR}}$: (a) W and C_0 , (b) Q and W , (c) C_0 and Q ; $q_{\max\text{MTZ}}$: (d) W and C_0 , (e) Q and W , (f) C_0 and Q . Red areas indicate maximum response, green shows minimum, and yellow represents intermediate values. The response is maximized in red zones.

Jurado-Davila et al. [50] found that during the evaluation of batch adsorption for the pharmaceuticals paracetamol and atenolol, both compounds exhibited rapid kinetics on activated carbon, reaching equilibrium at 60 min, with removal rates of 98% for PAR and 95% for ATL. The authors noted that prior to this time, the removal rates were approximately 50%, and after 60 min, the removal occurred at a much slower pace, with minimal changes in residual concentration over time. Similarly, de Oliveira et al. [41] observed that an increase in contact time resulted in higher pollutant removal rates and adsorption capacities, ultimately reaching process equilibrium. They attributed these results to the characteristics of the batch process, where initially, all active sites on the carbon activated surface are available facilitating rapid mass transfer. Over time, as these sites become occupied, their availability diminishes, leading to equilibrium.

After evaluating the influence of various parameters on the adsorption process, the optimal conditions for simultaneous pharmaceutical degradation were established. Based on the proposed design, an ideal solution pH of 7, a solid mass of 2.0 g, and a contact time of 60 min were selected, resulting in a removal rate of over 98.1% and an adsorption capacity of 4.0 mg/g for the pharmaceuticals. These findings underscore the significance of careful planning and optimization of process variables to achieve high removal rates and adsorption capacities in water or effluent treatment applications.

The results of the ANOVA analysis for the statistical model can be found in the Supplementary Material, specifically in Table S1. Significant variables and interactions were identified by examining the p -value at a 95% confidence level ($p < 0.05$), and these

are highlighted in red in the table. Additionally, the Pareto chart is presented in Figure S1 (Supplementary Material).

The analysis indicates that the pH parameter (x_1) does not show significant evidence for pharmaceutical removal or batch adsorption capacity, regardless of the conditions applied, which aligns with the observations in Figure 5. In contrast, solid mass (x_3) and contact time (x_2) demonstrate significant evidence for both response factors, particularly for Q_e . These findings are illustrated in the Pareto diagram, where the parameters that interfered with the system and exceeded the 95% confidence level were process time and solid mass (Figure S1, Supplementary Material). Under these conditions, the model effectively captured the relationship between the variables and the responses. Additionally, the values of the coefficient of determination (R^2) and the adjusted coefficient of determination (R^2_{Adj}) were $R^2 = 0.9754$; $R^2_{Adj} = 0.9569$ for %R and $R^2 = 0.8893$; $R^2_{Adj} = 0.8063$ for Q_e (mg/g), respectively, indicating strong agreement between actual and predicted values and further confirming the satisfactory relationship among the data.

To approximate the mathematical relationship of the experimental matrix, two second-order polynomial equations were derived: the first relates to removal efficiency (R%; Equation (12)), and the second pertains to adsorption capacity (Q_e ; Equation (13)). These equations are applicable solely under the conditions employed in this study and take into account only the significant factors involved in their formulation.

$$R\% = 41.47 + 1.19x_2 + 27.53x_3 \quad (12)$$

$$Q_e \left(\frac{\text{mg}}{\text{g}} \right) = 3.68 + 0.0051x_2 - 2.69x_3 + 0.95x_3^2 \quad (13)$$

where R% and Q_e (mg/g) are, respectively, the removal and adsorption capacity, x_2 is time (min), and x_3 is solid mass (g).

The experimental results obtained were $R\% = 98.1$, while the model predictions indicated $R\% = 167.93$ and $Q_e = 2.406 \text{ mg/g}$. Although the determination coefficients (R^2) for both equations show that the model performs well overall, the comparison between the experimental and predicted values suggests that the model does not adequately capture all the variables and interactions of the process. The overestimation of R% and the underestimation of Q_e indicate that there are experimental factors or interactions between the variables that are not properly represented by the model.

The significant difference between the experimental and predicted values for R% may be attributed to a limitation of the model in relation to the actual experimental conditions, such as nonlinear interactions or effects of unconsidered variables. For Q_e , the underestimation may reflect the complexity of the adsorption process, which is not fully addressed by the second-order polynomial model. Therefore, a revision of the model, including the consideration of additional variables or more complex interactions, is necessary to improve its accuracy. Furthermore, adjustments to the experimental design may be important to explore a wider range of factors that influence removal efficiency and adsorption capacity.

However, in addition to the statistical aspects, it is also essential to consider all the experimental factors that may influence the results. The experimental data obtained were as expected, showing the desired behavior and aligning with the theoretical foundations of adsorption. The removal efficiency and adsorption capacity observed confirm that, despite the model's limitations, the underlying processes of adsorption occur as predicted, validating the experimental approach adopted. The integration of experimental and theoretical aspects is crucial for the development of more accurate models and for the optimization of adsorption processes.

Based on the results presented, it was possible to conclude that the adsorption of paracetamol (PAR) on the GAC column is predominantly governed by specific interaction

mechanisms, including electrostatic forces, hydrophobic interactions, hydrogen bonding (dipole–dipole), n - π interactions, van der Waals forces, π - π interactions, and pore-filling processes [51,53–56]. These mechanisms align with the Langmuir isotherm and pseudo-second-order kinetics, indicating strong and specific interactions between PAR molecules and the adsorbent surface. In contrast, the adsorption of MTZ is primarily driven by physical processes, exhibiting a lower adsorption affinity on activated carbon [57,58]. The behavior aligns with the Freundlich isotherm and is characterized by pseudo-stationary kinetics, where the rate-limiting step is attributed to mass transfer resistance in the external film.

These findings highlight the distinct adsorption behaviors of PAR and MTZ, demonstrating the varying roles of specific interactions and mass transfer dynamics in the adsorption process. The study underscores the effectiveness of GAC for removing pharmaceutical and organic compounds, providing valuable insights into the adsorption mechanisms that govern fixed-bed column systems.

3.3. Fixed-Bed Column Adsorption Evaluation

Similar to batch adsorption, Table 5 presents the experimental conditions and results obtained for each run in the fixed-bed column experiments. The breakthrough curves for each run can be viewed in Figure S2 (Supplementary Material). The findings indicate that the highest maximum adsorption capacities for both pharmaceuticals ($q_{\max\text{PAR}} = 49.49$ mg/g and $q_{\max\text{MTZ}} = 43.64$ mg/g) were achieved in Experiment 1 ($C_0 = 30$ mg/L; $Q = 5$ mL/min; $W = 0.50$ g).

Table 5. Experimental matrix and responses for simultaneous adsorption of pharmaceuticals in a fixed-bed column.

Run Order	Independent Variables			Response	
	C_0 (mg/L)	W (g)	Q (mL/min)	$q_{\max\text{PAR}}$ (mg/g)	$q_{\max\text{MTZ}}$ (mg/g)
1 *	30	0.50	5	49.49	43.64
2	30	0.50	15	6.54	5.36
3	30	1.00	5	10.85	10.66
4	30	1.00	15	32.54	9.01
5	50	0.50	5	2.46	4.86
6	50	0.50	15	14.59	8.22
7	50	1.00	5	6.55	4.33
8	50	1.00	15	6.45	11.25
9 *	40	0.75	10	3.11	2.42
10 *	40	0.75	10	3.72	10.21
11 *	40	0.75	10	10.21	15.92

Legend: * optimized results.

To identify the optimal operational conditions, the simultaneous adsorption of the pharmaceuticals was analyzed. As shown in Table 5, most runs exhibited similar behavior, with values of maximum adsorption capacity in comparable orders of magnitude. Additionally, in many cases, the adsorption of paracetamol was slightly favored, resulting in a higher $q_{\max}$ compared to metronidazole. This suggests a stronger interaction between the carbon activated and paracetamol.

Nadour et al. [59] studied the removal of paracetamol, metronidazole, and diclofenac using carbon-polymeric membranes, finding that diclofenac removal reached 44%, while paracetamol and metronidazole had removal rates of 4% and 8%, respectively. These differences can be attributed to variations in molecular weight and solubility among the compounds. The authors also noted that the removal efficiency varied according to the

concentrations of the compounds applied in the process, particularly for paracetamol and metronidazole, a trend observed in the present study as well. Additionally, they found that incorporating activated carbon into the membranes significantly increased removal rates by 34% for paracetamol and 28% for metronidazole, helping to explain the higher values recorded in this work.

Bouarroudj et al. [60] compared three removal methods for metronidazole and paracetamol in a batch process, observing that photocatalysis had a significant impact on the photodegradation of both compounds, while adsorption and photolysis were less effective. This indicates that achieving effective adsorption of these compounds can be challenging using this method. However, the present study achieved high adsorption efficiencies with continuous adsorption, recording $q_{\max}$ values of 49.49 mg/g for paracetamol and 43.64 mg/g for metronidazole.

Comparing the results with the adsorption of only PAR in batch process, also using GAC, by Haro et al. [61], the authors obtained $q_{\max\text{PAR}} = 31.5$ mg/g, and Ilavský et al. [62] evaluated the adsorption of specific pharmaceuticals (paracetamol, carbamazepine, metronidazole, and caffeine) in a single and batch approach and observed that the removal capacity for PAR ranged between 0.071 and 0.434 $\mu\text{g/g}$, while for MTZ it was 0.11 to 0.539 $\mu\text{g/g}$ using different activated carbon materials, showing a high efficiency of removal rate in this study when applying the continuous adsorption, an industrial approach, even in a multi-compound system. In addition, through the batch experiments, also applied in this work, in item 3.2, the removal efficiencies were between 71% and 98%, showing that the operational conditions applied in the process significantly impact on the feasibility of the technique by simultaneously treating the adsorption of these compounds.

Thus, it can be concluded from the results that the best adsorption capacities were obtained at the lowest concentration of the pharmaceuticals, suggesting a faster saturation of the solid when the concentration is higher, also observed by other works reported in the literature [55,58,60]. Therefore, the best conditions obtained in the present work refer to Experiment 1, at lower concentration of pollutants, flow rate, and bed height, in which, under these conditions, the process control is facilitated, and there is a smaller amount of adsorbent required, resulting in the minimization of waste generation and lower process costs.

The analysis of the breakthrough curves (Figure S2, Supplementary Material) revealed that all curves deviate from the ideal sigmoid (S) shape characteristic of breakthrough curves. The S-shape indicates a vertical alignment of the adsorbed species and is associated with cooperative adsorption. When the breakthrough curve deviates from this format, traditional models commonly used in the literature (such as Thomas, Yan, and Yoon-Nelson) often fail to accurately predict the outcomes due to their inability to account for the curve's asymmetry. This limitation can be addressed by using the Gompertz models [61,62], which were used in this study, proposed by Hu et al. [40] for multicomponent adsorption systems.

It is noteworthy that, as observed in Figure S2, some curves do not reach the saturation point, defined as when 95% of the initial compound concentration is achieved. This is particularly true for the lower concentration mixtures ($C_0 = 30$ mg/L and 40 mg/L), including the optimal point. Additionally, better column performance is indicated by more favorable parameters, resulting in longer saturation times [50]. This suggests that, for industrial applications, the adsorbent can be utilized for extended periods before requiring replacement or regeneration. It is significant to note that the optimal point corresponds to the lowest adsorbent mass used in this study ($W = 0.5$ g). It is worth mentioning that Haro et al. [61] also investigated the removal of paracetamol in a fixed-bed column system, finding that the best operating condition was achieved with 0.5 g of GAC, resulting in extended saturation times for the column. They pointed out that increasing the bed

mass indefinitely would eventually limit operation due to dispersion effects caused by the increased interstitial spaces within the adsorbent [63].

In Figure 5, the surface area and contour plots for simultaneous pharmaceutical removal by fixed-bed column adsorption are presented, depicting the response factors studied in this work ($q_{\max\text{PAR}}$ (mg/g) and $q_{\max\text{MTZ}}$ (mg/g)).

According to Figure 5, the bed height significantly influences the response factors evaluated, particularly in relation to the process flow rate. While some authors in the literature have indicated that a greater bed height increases the total surface area of the adsorbent [50], the adsorption capacity for pharmaceuticals tends to decrease with higher mass. For the conditions assessed, it appears that smaller masses combined with lower flow rates are adequate to achieve optimal adsorption capacity (Figure 5b,e).

This behavior can be attributed to three primary factors: (1) not all binding sites may be accessible to adsorbate molecules at greater bed heights, potentially due to overlapping active sites, which leads to the underutilization of certain particles [64]; (2) the channeling effect, where the solution flows preferentially near the column wall where there is more void space [65]; (3) the axial dispersion phenomenon, which can dominate during mass transfer and hinder solute diffusion into the adsorbent pores [66]. Therefore, a shallower bed depth has proven more advantageous for achieving optimal pharmaceutical adsorption capacity.

The initial concentration of pharmaceuticals has a proportional effect on the mass present in the process. Lower concentrations of pharmaceuticals at a shallower bed height (Figure 5a,d) were sufficient to yield favorable process responses without the need for higher concentrations. A similar trend was noted for the flow rate (Figure 5c,f). This was expected, as an increase in feed flow rate reduces the contact time between the adsorbate and the adsorbent, which in turn decreases bed capacity [67,68]. Scheufele et al. [69] observed that while increasing the initial pollutant concentration generally enhances adsorption capacity, it can also reduce process efficiency (bed utilization). This finding aligns with the present study, where higher concentrations of pharmaceuticals led to decreased process efficiency.

An ANOVA analysis and a Pareto diagram were conducted for the two response factors to assess the significance of the variables applied in the process (see Table S2 and Figure S3 in the Supplementary Material). The validity of the polynomial model was evaluated by calculating the coefficient of determination (R^2). By examining the Pareto chart, one can determine the magnitude and importance of the effects within the process.

This analysis revealed that only the $q_{\max\text{PAR}}$ response factor was influenced by the evaluated variables in the process, specifically the initial concentration (C_0) and the interaction between the solid mass (W) and flow rate (Q). The R^2 and adjusted R^2 values for this factor were 0.5297 and 0.0, respectively. In contrast, for the $q_{\max\text{MTZ}}$ response factor, although no significant variables were identified, it yielded R^2 and adjusted R^2 values of 0.811 and 0.526, respectively. These values raise questions about the reliability of the relationship between predicted and experimental data, and in some cases, they may be considered statistically unsatisfactory. Furthermore, when investigating each parameter and their interactions individually (Figure S3), it was noted that parameters with negative coefficients corresponded to undesirable or antagonistic effects, while those with positive coefficients indicated desirable and/or synergistic effects on the efficiency of the adsorption process.

To confirm the previous statements and approximate the mathematical relationship of the experimental matrix, as in batch adsorption, two second-order polynomial equations were obtained. The first equation refers to $q_{\max\text{PAR}}$ (Equation (14)), and the second equation

to $q_{\max\text{MTZ}}$ (Equation (15)). These equations are valid only for the conditions applied in the present study, and only the significant factors were considered.

$$q_{\max\text{PAR}}\left(\frac{\text{mg}}{\text{g}}\right) = 112.827 - 1.8604x_1 - 5.5251x_3 + 1.3102x_2x_3 \quad (14)$$

$$q_{\max\text{MTZ}}\left(\frac{\text{mg}}{\text{g}}\right) = 133.046 \quad (15)$$

where $q_{\max\text{PAR}}$ and $q_{\max\text{MTZ}}$ (mg/g) represent the maximum adsorption capacity for paracetamol and metronidazole, respectively; x_1 is the initial concentration of the pharmaceuticals; x_2 represents the mass (W; g); and x_3 is the feed flow rate (Q; mL/min).

Considering these equations and the optimal experimental conditions obtained through the experimental procedure, it can be observed that the values of $q_{\max\text{PAR}} = 32.665$ mg/g and $q_{\max\text{MTZ}} = 133.046$ mg/g are derived from the model, with the latter showing that the variables involved do not significantly influence the adsorption of the compound. When analyzing these results in conjunction with the R^2 value obtained in this study, it becomes evident that the experimental design and the applied model do not statistically support the proposed experiment. The R^2 value indicates that the model does not account for a sufficient amount of variability in the data, meaning that the relationship between the factors and the response ($q_{\max}$ of the compounds) may not be accurately represented by the current model. Additionally, the experimental design, while based on established methodologies, may not fully capture the complexity or the interactions of the variables, leading to less reliable conclusions. Therefore, a reassessment of the model or potential refinement in the experimental design is necessary to improve the statistical robustness and ensure that the findings are both valid and reliable. However, since only significant factors were considered in the model, the difference in the values of $q_{\max}$ also may be attributed to the conditions that were not statistically significant, which are inherently included in the experimental setup.

Even though the interaction between W and Q is statistically significant only for the paracetamol ($q_{\max\text{PAR}}$), as presented, a similar phenomenon is observed when considering metronidazole (Figure 5d–f). This insight underscores the complexity of multicomponent adsorption phenomena and emphasizes the need for tailored experimental and modeling approaches to effectively characterize and predict these interactions.

Table 6 shows the comparison of the maximum adsorption capacity of organic molecules at different adsorbents and GAC present in the literature.

Table 6. Comparison of the maximum adsorption capacity ($q_{\max}$) of organic molecules at 25 °C in GAC present in the literature.

Compound	$Q_{\max}$ (mg/g)	Reference
Paracetamol	49.5	Present study
Metronidazole	43.6	Present study
Atenolol	79.9	Sotelo et al. [70]
Benzalkonium chloride	26.3	Costa et al. [52]
Caffeine	7.3	de Oliveira et al. [63]
Tetracycline	2.08	Toffoli de Oliveira et al. [71]
Carbamazepine	56.1	Almuntashiri et al. [72]
Metronidazole	32.2	Almuntashiri et al. [72]
Salicylic acid	205.1	Bernal et al. [73]
Isoproturon	33.2	Sotelo et al. [70]

The results in Table 6 highlight significant variations in the maximum adsorption capacity ($q_{\max}$) of organic compounds on GAC, depending on the compound's physicochemical properties and specific interactions with the adsorbent surface. In this study, paracetamol (49.5 mg/g) and metronidazole (43.6 mg/g) showed moderate adsorption capacities compared to other compounds. For instance, salicylic acid (205.1 mg/g) [70] exhibited much higher $q_{\max}$, likely due to stronger hydrophobic or electronic interactions with GAC, while tetracycline (2.08 mg/g) [74] had the lowest capacity, indicating less favorable interactions. Compounds such as atenolol (79.9 mg/g) [75] and carbamazepine (56.1 mg/g) [69] displayed adsorption capacities closer to the values observed in this study. These findings emphasize that GAC interactions with organic compounds depend on factors like polarity, functional groups, molecular size, and hydrophobicity.

Mathematical Modeling of Breakthrough Curves

Table 7 presents the parameter values and statistical metrics derived using the MCMC method for the adsorption of paracetamol and metronidazole, respectively, under the optimal operational conditions (run order 1). The values presented in Table 7 are the posteriori means of the parameters, containing the lower and upper limits, according to the confidence interval applied (99%) in the present study.

Table 7. Values for the model parameters and the statistical metrics by the MCMC method for the multicomponent adsorption of Experiment 1 ($C_0 = 30$ mg/L, $Q = 5$ mL/min and $W = 0.5$ g).

Model			
Logistic		Gompertz	
Parameters and Units	Values Estimated by MCMC	Parameters and Units	Values Estimated by MCMC
Paracetamol			
c	65.19 (60.33–68.86)	c	56.19 (53.62–59.05)
k (1/min)	0.0075 (0.0071–0.0078)	k (1/min)	0.0045 (0.0044–0.0046)
τ (min)	292.44 (288.27–296.10)	τ (min)	205.43 (201.32–209.89)
k^* (1/min)	0.019 (0.018–0.020)	k^* (1/min)	0.0165 (0.016–0.017)
τ^* (min)	90.52 (89.01–91.91)	τ^* (min)	67.36 (66.30–68.39)
R^2	0.708	R^2	0.754
R^2_{ajstd}	0.686	R^2_{ajstd}	0.736
AIC	1653.10	AIC	1301.10
AIC _C	1653.20	AIC _C	1301.20
BIC	1654.20	BIC	1302.20
Metronidazole			
c	69.09 (65.75–73.27)	c	59.27 (56.98–61.63)
k (1/min)	0.0083 (0.0080–0.0086)	k (1/min)	0.0053 (0.0051–0.0054)
τ (min)	249.79 (246.08–253.87)	τ (min)	168.82 (165.14–172.68)
k^* (1/min)	0.020 (0.019–0.021)	k^* (1/min)	0.018 (0.017–0.019)
τ^* (min)	85.99 (84.51–87.42)	τ^* (min)	63.37 (62.39–64.37)
R^2	0.714	R^2	0.760
R^2_{ajstd}	0.692	R^2_{ajstd}	0.742
AIC	1537.60	AIC	1207.10
AIC _C	1537.70	AIC _C	1207.20
BIC	1538.80	BIC	1208.30

Figure 6 illustrates the breakthrough curve modeling for the simultaneous adsorption of the two pharmaceuticals in a fixed bed column during Experiment 1, with the Logistic and Gompertz models applied to the multicomponent adsorption system.

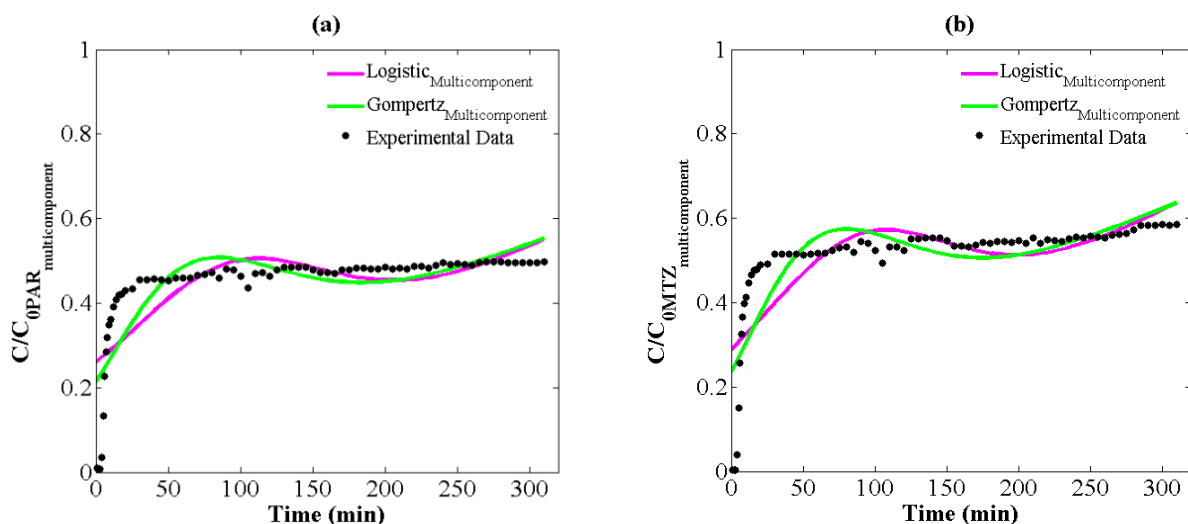


Figure 6. Breakthrough curve modeling of (a) paracetamol and (b) metronidazole adsorption in fixed bed column for Experiment 1 for multicomponent models (conditions: $C_0 = 30$ mg/L, $Q = 5$ mL/min and $W = 0.5$ g).

From Table 7 it is observed that the Gompertz model for multicomponent adsorption provided the best fit for both compounds, with the following statistical metrics: PAR: $R^2 = 0.754$, adjusted $R^2 = 0.736$, $AIC = 1301.10$, $AIC_C = 1301.20$, and $BIC = 1302.20$; MTZ: $R^2 = 0.760$, adjusted $R^2 = 0.742$, $AIC = 1207.10$, $AIC_C = 1207.20$, and $BIC = 1208.30$. The logistic model represents a more symmetric and idealized breakthrough curve, making it suitable for systems with an even progression from low to high concentration. In contrast, the Gompertz model is more flexible and better suited for systems with delayed breakthroughs, where the transition from low to high concentration is asymmetric.

In this study, the Gompertz model was found to better represent the experimental data due to the pronounced asymmetry observed in the breakthrough curves. Furthermore, as shown in Figure 5, the rapid transition from low to high concentrations in the experimental data aligns more closely with the estimates produced by the MCMC technique for the Gompertz model. To verify the convergence of the posterior distributions and confirm the viability of the Gompertz model, the MCMC chains for both paracetamol and metronidazole are presented in Figures 7 and 8, respectively.

From Figures 7 and 8 it is possible to observe the convergence of the Markov chains for all parameters estimated from both models through Bayesian inference. This result indicates that the technique is applicable to multicomponent adsorption systems, since the method was able to adequately explore the parameter space. Thus, the MCMC technique is capable of dealing with the complexity of the modeled system, which presents complex interactions between parameters, reinforcing the reliability of the estimated values and allowing the assessment of uncertainties associated with the parameters, being essential to validate the model in complex experimental conditions.

These results reinforce the conclusion that the methodology employed in this study, alongside with practical models for multicomponent adsorption systems, is advantageous for scaling up the implementation of the processes. The use of asymmetric models derived from Gompertz [72] can further enhance the prediction of experimental data, supporting their application in various adsorption processes for practical implementation. In addition, Hu et al. [40] also concluded that the Gompertz model is superior to the logistic model in terms of the fitting accuracy.

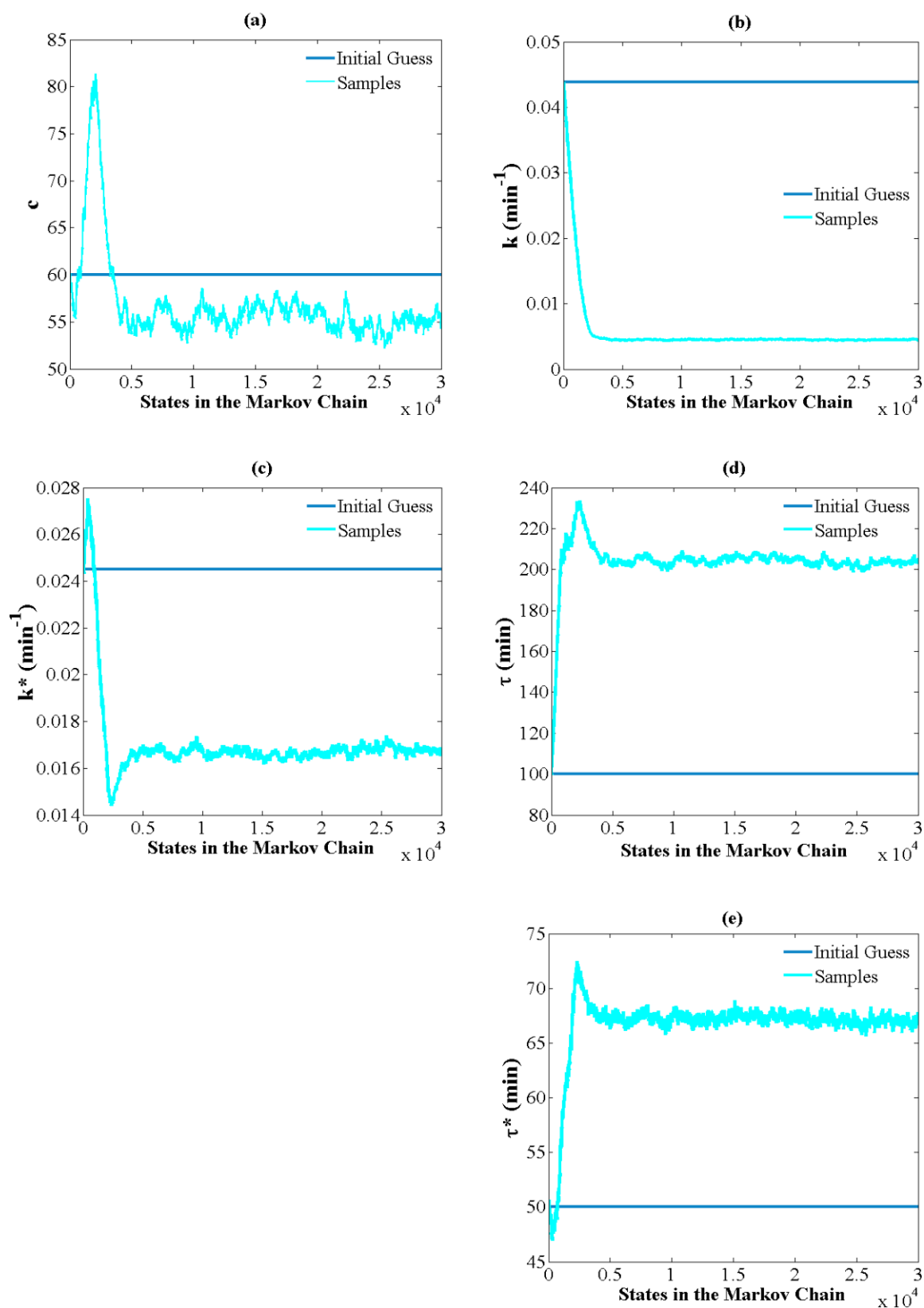


Figure 7. Markov chains of the parameters of the Gompertz model, for the adsorption of paracetamol for Experiment 1: (a) c ; (b) k ; (c) k^* ; (d) τ ; and (e) τ^* (conditions: $C_0 = 30$ mg/L, $Q = 5$ mL/min and $W = 0.5$ g).

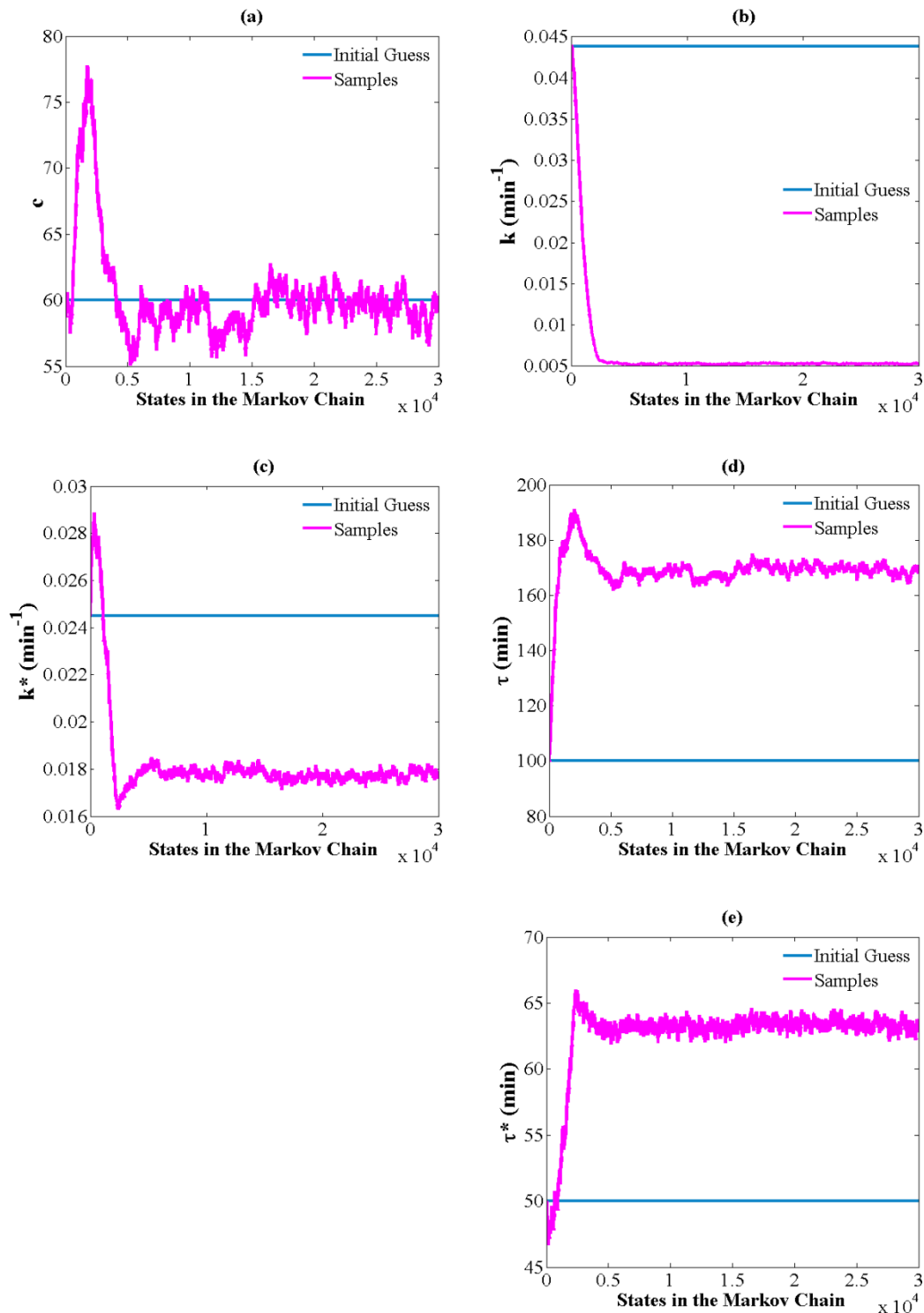


Figure 8. Markov chains of the parameters of the Gompertz model, for the adsorption of metronidazole for Experiment 1: (a) c ; (b) k ; (c) k^* ; (d) τ ; and (e) τ^* (conditions: $C_0 = 30$ mg/L, $Q = 5$ mL/min and $W = 0.5$ g).

Hu et al. [40] observed that, in multicomponent adsorption systems, the concentration of a compound in the treated effluent may exceed its initial concentration in the feed effluent. The authors attributed this phenomenon to the removal of strongly binding ions in the upper sections of the column, which consequently leads to the release of an equivalent amount of weaker ions. However, in the present study, the typical behavior of multicomponent adsorption was not observed. This discrepancy could be attributed to

differences in the adsorption dynamics resulting from the specific experimental conditions used. Longer process times may be required to observe this effect fully. Additionally, variations in the feed composition and/or the adsorption properties of the adsorbent could prevent the displacement of weaker ions by those with a higher affinity for the adsorbent. These findings emphasize that each multicomponent system requires a careful evaluation of its unique process dynamics.

It is important to note that the parameters of Gompertz models do not provide specific insights into the dominant mechanisms of the adsorption process, as the model was originally developed for different applications. However, Juella et al. [65] observed that the parameters vary with changes in the operational conditions of the fixed-bed column, making the model particularly useful for scale-up studies. Hu et al. [40] concluded that the parameters c , k , τ , k^* , and τ^* are essential for tailoring the logistic and Gompertz models to accurately describe adsorption processes. They allow for the adjustment of curve steepness, position, and interaction effects, making the models applicable to multicomponent systems. The observed asymmetry in breakthrough curves arises from differences in adsorption mechanisms, reactant reactivity, and intraparticle diffusion, highlighting the complexity of real-world adsorption systems.

By comparing the results by Hu et al. [40] with the ones found in this work is possible to conclude that the parameter's values are in the same order of magnitude for binary fixed-bed adsorption, in which a higher k value implies a steeper curve, indicating more favorable adsorption and τ shifts the position of the breakthrough curve along the time axis, aligning with the experimental data. However, it is important to note that the model adjustments to the experimental data yielded $R^2 \approx 0.70$ for both compounds and models. This moderate fit may be attributed to the fact that the breakthrough curves only reach approximately 50–60% of the initial pharmaceutical concentrations. Despite this, the methodology demonstrates its suitability for modeling multicomponent fixed-bed adsorption systems effectively.

4. Engineering Implications

Adsorption for the removal of multi-component pharmaceuticals holds great potential for treating wastewater contaminated with pharmaceutical substances, with promising large-scale applicability. To ensure the operational and economic viability of this technology, it is crucial to optimize parameters such as flow rate, adsorbent mass, and initial pollutant concentration, along with strict control of the process variables. The scalability will depend on the ability to maintain consistent performance under real operational conditions. In multi-component systems, competition among pollutants for adsorption sites is a challenge, as interactions between pharmaceuticals can compromise process efficiency. A detailed understanding of these interactions is essential. Solutions such as selective modification of the adsorbent or the use of functionalized materials can be adopted to improve process efficiency.

Although the primary focus of the study has been on the technical understanding of the adsorption process, the economic feasibility of implementing a fixed-bed system for pharmaceutical pollutant removal should be considered in future research. The cost of adsorbents, along with waste generation, can pose limiting factors in large-scale operations. In this context, regeneration and reuse of the adsorbent are important strategies to reduce overall operational costs and improve sustainability. Although regeneration tests were not performed in the present study, their evaluation is strongly recommended in future work, especially to assess long-term viability under real application scenarios. Among the possible regeneration strategies, thermal regeneration, chemical regeneration (e.g., using alkaline or acidic solutions), and solvent washing stand out as effective techniques to restore adsorbent performance. The choice of regeneration method must consider not only efficiency but also

environmental impact, in line with the principles of green chemistry. These methods, when optimized, can significantly extend the service life of the adsorbent and reduce the need for raw material consumption, contributing to a more sustainable process. Additionally, in cases where regeneration is not feasible, environmentally responsible disposal methods should be adopted to avoid secondary contamination.

The use of alternative and sustainable materials, or adsorbent regeneration, can help reduce operational costs. However, the processing costs of these materials, along with a life cycle assessment (LCA), are essential for evaluating the environmental impact and identifying opportunities to optimize resource use. Additionally, operational costs, such as the energy consumption required to maintain flow rates and maintenance and monitoring costs, should be considered to assess the long-term feasibility of the process.

Future investigations may address these issues and explore the use of alternative adsorption materials, offering more sustainable and cost-effective solutions. The development of real-time monitoring and process automation technologies could also improve control and operational efficiency, optimizing performance and reducing operational costs.

Finally, although the results of this study are promising, adapting adsorption technology for pharmaceutical removal in industrial wastewater treatment plants will require consideration of operational limitations, such as available space, waste management, and integration with existing infrastructure. The successful large-scale implementation of this technology will depend on modular and scalable solutions, as well as partnerships among academia, industry, and regulatory agencies.

5. Conclusions

In this study, the influence of parameters on the adsorption process was investigated through batch-scale experiments, where the optimal conditions for the multicomponent removal of the compounds PAR and MTZ were determined to be pH 7, a mass of 2.0 g, and 60 min of contact time, achieving a removal efficiency of 98.1% for both compounds. Fixed-bed column experiments were also conducted to explore the behavior of pharmaceutical removal processes. It was observed that the initial pharmaceuticals concentration applied to the column significantly affected the response factor of maximum adsorption capacity for paracetamol, followed by the interaction between bed height and flow rate. Although the interaction was not statistically significant for MTZ, a similar trend was observed for this compound. The optimal operating conditions for this stage were determined as $C_0 = 50$ mg/L, $Q = 5$ mL/min, and $W = 0.5$ g. These conditions resulted in maximum adsorption capacities of 49.49 mg/g and 43.64 mg/g for PAR and MTZ, respectively, and the resulted breakthrough curves were asymmetric. Within the application of the Bayesian technique, the multicomponent Gompertz model was selected as the most suitable for representing experimental measurements and can be used to scale-up the process. Thus, the mathematical modeling of the breakthrough curve, applied using Bayesian techniques, has proven to be a capable method for dealing with the complexity of multicomponent systems, which present complex interactions between parameters, reinforcing the reliability of the estimated adsorption parameters. In addition, these findings underscore the complexity of multicomponent adsorption phenomena and emphasize the necessity for experimental and modeling approaches to accurately characterize and predict these interactions.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/app15137316/s1>: Table S1. ANOVA Table for the Variables: (i) Removal and (ii) Adsorption Capacity; Figure S1. Pareto diagrams for simultaneous pharmaceuticals batch adsorption (A, removal is the response variable; B; Q_e is the response variable); Figure S2. Breakthrough Curves Obtained by Simultaneous Adsorption of Pharmaceuticals in a Fixed-Bed Column; Table S2. ANOVA Table for the Response Variables: (i) $q_{\max\text{PAR}}$ and (ii) $q_{\max\text{MTZ}}$

of Adsorption in Fixed-Bed Column*; Figure S3. Pareto diagrams for simultaneous pharmaceuticals removal by fixed-bed column adsorption (A, $q_{\max\text{PAR}}$ is the response variable; B; $q_{\max\text{MTZ}}$ is the response variable)*.

Author Contributions: L.R.d.C.C.: Conceptualization, Methodology, Software, Validation, Investigation, Writing—Original Draft, Data Curation; J.T.d.O.: Conceptualization, Methodology, Software, Data Curation, Writing—Original Draft, Writing—Review and Editing; F.S.S.: Methodology, Formal Analysis; L.A.F.: Writing—Review and Editing, Supervision. All authors have read and agreed to the published version of the manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

PAR	Paracetamol
MTZ	Metronidazole
pH	Potential of Hydrogen
XRD	X-ray Diffraction
BJH	Barrett–Joyner–Halen
BET	Brunauer–Emmett–Teller
FTIR	Fourier-Transform Infrared Spectroscopy
BBK	Box–Behnken Design
ANOVA	Analysis of Variance
RSM	Response Surface Methodology
GAC	Granular Activated Carbon
AIC	Akaike Information Criterion
AIC _C	Corrected Akaike Information Criterion
BIC	Bayesian Information Criterion
MCMC	Markov Chain Monte Carlo
MH	Metropolis–Hastings Algorithm
IUPAC	International Union of Pure and Applied Chemistry
LCA	Life Cycle Assessment

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